BSCS FINAL PROJECT

Software Requirements Specification
Phase 1

SAFORA (Smart Ride)

Project Advisor

Tanveer Ahmed

Presented by:

Group ID: 11

S3F22UBSCS081	Mirza Umer Ikram
S3F22UBSCS088	Ruhma Bilal

Faculty of Information Technology & Computer Science

University of Central Punjab

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1. INTRODUCTION

1.1 Purpose

This Software Requirements Specification (SRS) document provides a complete description of the SAFORA (Smart Ride) system - an AI-powered ride-hailing platform implementing proactive safety measures. This document is intended for:

- Development team members
- Project stakeholders and supervisors
- Quality assurance personnel
- Future maintenance teams

The document follows IEEE Std 830-1998 guidelines for software requirements specifications.

1.2 Scope

Product Name: SAFORA (Smart Ride)

Product Description: SAFORA is an intelligent ride-hailing platform that addresses critical safety and efficiency gaps in existing transportation services through AI-driven proactive safety mechanisms.

1.2.1 Product Features

1. Pink Pass (Biometric Women-Only Mode)

- Two-layer verification using CNIC and liveness detection
- Haar Cascade for face detection
- CNN-based blink verification for anti-spoofing
- Verification completion in <15 seconds

2. Safety Sentinel (Automated Route Monitoring)

- Real-time GPS tracking and route deviation detection
- Point-to-line distance calculation using Turf.js

- Automated alerts when deviation >500m for >30 seconds
- Dual notification system (admin dashboard + SMS to emergency contacts)

3. Smart Driver Selection

- AI-driven matching algorithm: Score = 0.50*Distance + 0.30*Rating + 0.20*Fairness
- Fairness metric: IdleTime/TotalOnlineTime (last 4 hours)
- Reduces driver monopolization and wait times

1.2.2 Benefits

For Passengers: Enhanced safety through proactive monitoring, verified women-only rides, faster pickup times

For Drivers: Fair ride distribution, reduced idle time, transparent rating system

For Administrators: Real-time monitoring, automated alerts, data-driven insights

1.2.3 In Scope

- User registration and authentication (Passenger/Driver/Admin)
- Ride booking with real-time price estimation
- GPS-based tracking and route monitoring
- Pink Pass enrollment and verification
- Automated safety alerts
- Smart driver matching algorithm
- Admin dashboard with analytics
- Cash payment processing

1.2.4 Out of Scope

- Multi-city operations (Phase 1 limited to Lahore)
- Third-party payment gateway integration
- Carpooling/ride-sharing features
- Advanced fraud detection systems
- International operations

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
CNIC	Computerized National Identity Card
EAR	Eye Aspect Ratio
GPS	Global Positioning System
JWT	JSON Web Token
MERN	MongoDB, Express.js, React, Node.js
NLP	Natural Language Processing
PKR	Pakistani Rupees
REST	Representational State Transfer
SMS	Short Message Service
SOS	Emergency distress signal
SRS	Software Requirements Specification
TPR	True Positive Rate
FPR	False Positive Rate

1.4 References

1. [1] IEEE Std 830-1998, "IEEE Recommended Practice for Software Requirements Specifications"
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1.5 Overview

This SRS document is organized as follows:

- Section 2 provides an overall description of the SAFORA system, including product perspective, functions, user characteristics, and constraints.
- Section 3 details specific functional and non-functional requirements.
- Section 4 presents system models including use case, sequence, class, and ER diagrams.

- Section 5 outlines the project plan with methodology, timeline, and resource allocation.
- Section 6 contains appendices with supplementary information.

2. OVERALL DESCRIPTION

2.1 Product Perspective

SAFORA is a new, self-contained ride-hailing platform designed to address specific safety and efficiency gaps in existing services operating in Pakistan. The system consists of four main components:

2.1.1 System Architecture

Backend Server (Node.js/Express)

- RESTful API for all client applications
- Real-time communication via Socket.io
- MongoDB database for data persistence
- JWT-based authentication

AI Microservice (Python/Flask)

- Pink Pass liveness detection
- Price prediction using Linear Regression
- Smart driver matching algorithm
- Sentiment analysis and demand clustering

Mobile Applications (React Native)

- Passenger app for iOS and Android
- Driver app for iOS and Android
- Real-time GPS tracking
- In-app notifications

Admin Dashboard (React.js)

- Web-based monitoring interface
- Real-time safety alerts
- Analytics and reporting
- Heatmap visualization

2.1.2 System Interfaces

Hardware Interfaces:

- Mobile device GPS sensors
- Mobile device cameras (for Pink Pass verification)
- Server infrastructure (cloud or on-premise)

Software Interfaces:

- Google Maps API for routing and geocoding
- Twilio API for SMS notifications
- MongoDB database (v6.0+)
- Node.js runtime (v18+)
- Python runtime (v3.9+)

Communication Interfaces:

- HTTPS for secure API communication
- WebSocket (Socket.io) for real-time updates
- SMS for emergency notifications

2.2 Product Functions

Function	Description	Priority
User Management	Registration, authentication, profile management for all user roles	High
Pink Pass	Biometric verification for women-only rides using liveness detection	High
Ride Booking	Request rides with price estimation and driver matching	High
Safety Sentinel	Automated route monitoring with deviation detection	Critical
Driver Management	Driver registration, status tracking, fairness metrics	High
Admin Dashboard	Real-time monitoring, alerts, analytics, and reporting	Medium

2.3 User Characteristics

User Type	Technical Expertise	Primary Needs	Usage Frequency
Passengers	Basic smartphone usage	Safe, affordable, reliable transportation	Daily to weekly
Drivers	Basic smartphone and navigation	Fair ride distribution, consistent income	Daily
Administrators	Moderate to advanced	Monitor safety, manage users, analyze data	Daily

2.4 Constraints

2.4.1 Regulatory Constraints

- Must comply with Pakistan's data protection regulations
- CNIC verification required for all users
- SMS notifications must follow PTCL guidelines

2.4.2 Technical Constraints

- Mobile apps must support Android 8.0+ and iOS 13.0+
- Backend must handle 1000+ concurrent users
- API response time must be <500ms for 95% of requests
- GPS accuracy must be within 10 meters

2.4.3 Business Constraints

- Phase 1 limited to Lahore city
- Cash-only payments initially

- Development timeline: 22 weeks
- Budget constraints for cloud infrastructure

2.5 Assumptions and Dependencies

2.5.1 Assumptions

- Users have smartphones with GPS and camera capabilities
- Reliable internet connectivity (3G/4G/WiFi) available
- Google Maps API will remain accessible and affordable
- Twilio SMS service will maintain 99% uptime

2.5.2 Dependencies

- MongoDB database availability
- Third-party API services (Google Maps, Twilio)
- Mobile OS platform stability
- Cloud infrastructure reliability

3. SPECIFIC REQUIREMENTS

3.1 Functional Requirements

ID	Requirement	Priority
FR-01	System shall allow user registration with CNIC verification	High
FR-02	System shall authenticate users using JWT tokens	High
FR-03	System shall allow Pink Pass enrollment with video upload	High
FR-04	System shall detect faces using Haar Cascade (>95% accuracy)	High
FR-05	System shall verify liveness using CNN (>98% accuracy, <5% FPR)	Critical
FR-06	System shall complete Pink Pass verification within 15 seconds	High
FR-07	System shall allow ride requests with pickup/dropoff locations	High
FR-08	System shall predict ride price using AI (+/- PKR 30 accuracy)	High
FR-09	System shall match drivers using weighted algorithm	High
FR-10	System shall monitor rides with GPS tracking (5-second intervals)	Critical
FR-11	System shall detect route deviation (>500m for >30 seconds)	Critical
FR-12	System shall send alerts to admin and emergency contacts within 30s	Critical
FR-13	System shall track driver online/offline status and location	High
FR-14	System shall calculate driver fairness metric	Medium
FR-15	System shall display real-time safety alerts on admin dashboard	High

3.2 Non-Functional Requirements

Category	Requirement	Metric
Performance	API response time	<500ms for 95% requests
Performance	Pink Pass verification time	<15 seconds
Performance	Concurrent users support	1000+ users
Security	Password hashing	bcrypt (cost factor 10)
Security	CNIC encryption	AES-256
Security	API communication	HTTPS/TLS 1.3
Reliability	System uptime	99% during business hours
Reliability	Safety features availability	99.9%
Usability	Minimum font size	14px
Usability	Language support	Urdu and English
Maintainability	Code coverage	>80% for unit tests
Maintainability	Code standards	ESLint and PEP 8

3.3 Interface Requirements

3.3.1 User Interfaces

Mobile App Screens:

7. Authentication (Login/Register)
8. Home/Dashboard
9. Ride Booking
10. Pink Pass Enrollment
11. Ride Tracking
12. Ride History
13. Profile Management
14. Emergency SOS

Admin Dashboard Pages:

15. Real-time Monitoring
16. Safety Alerts
17. User Management
18. Analytics & Reports
19. Heatmap Visualization

3.3.2 Hardware Interfaces

- GPS sensor for location tracking
- Camera for Pink Pass verification
- Network interface for API communication

3.3.3 Software Interfaces

- Google Maps API v3 for routing and geocoding
- Twilio REST API for SMS notifications
- MongoDB database driver
- Socket.io for WebSocket communication

4. SYSTEM MODELS

4.1 Use Case Diagrams

The system includes use cases for three primary actors: Passengers, Drivers, and Administrators.

Passenger Use Cases:

- Register Account
- Login
- Enroll in Pink Pass
- Request Ride
- Track Ride
- Rate Driver
- Trigger SOS

Driver Use Cases:

- Register as Driver
- Update Availability
- Accept/Reject Ride
- Navigate to Pickup
- Complete Ride
- View Earnings

Admin Use Cases:

- Monitor Active Rides
- View Safety Alerts
- Manage Users
- Generate Reports
- Analyze Heatmaps

4.2 Sequence Diagrams

4.2.1 Ride Request Flow

20. 1. Passenger selects pickup/dropoff locations
21. 2. System calculates route and price
22. 3. System matches optimal driver
23. 4. Driver receives notification
24. 5. Driver accepts ride
25. 6. Passenger receives driver details
26. 7. Ride begins with GPS tracking

4.2.2 Safety Alert Flow

27. 1. Safety Sentinel detects route deviation
28. 2. System creates alert record
29. 3. Admin dashboard receives Socket.io notification
30. 4. Emergency contacts receive SMS
31. 5. Admin reviews and resolves alert

4.3 Class Diagrams

Class Name	Attributes	Methods
User	id, name, email, phone, password, role	register(), login(), updateProfile()
Passenger	emergencyContacts, pinkPassVerified	enrollPinkPass(), requestRide()
Driver	licenseNumber, vehicleInfo, rating, location	updateStatus(), acceptRide()
Ride	passenger, driver, pickup, dropoff, status	create(), updateStatus(), complete()
Alert	ride, type, severity, location	create(), notify(), resolve()

4.4 Entity Relationship Diagrams

Entity	Attributes	Relationships
Users	_id, name, email, phone, cnic, gender, role	One-to-One with Drivers
Drivers	_id, user, license, vehicle, rating, location	One-to-Many with Rides
Rides	_id, passenger, driver, pickup, dropoff, status	Many-to-One with Passengers/Drivers
Alerts	_id, ride, type, severity, location	Many-to-One with Rides

5. PROJECT PLAN

5.1 Development Methodology

Methodology: Agile Scrum with 2-week sprints

Rationale: Agile allows iterative development, frequent testing, and flexibility to adapt to changing requirements.

Team Roles:

- Mirza Umer Ikram: Backend Development, AI Services
- Ruhma Bilal: Frontend Development, UI/UX Design
- Tanveer Ahmed: Project Advisor, Technical Guidance

5.2 Timeline and Milestones

Phase	Duration	Tasks	Status
Phase 1: Setup	Weeks 1-2	Project structure, environment setup, SRS	Completed
Phase 2: Backend	Weeks 3-5	Express server, MongoDB, authentication, APIs	Completed
Phase 3: AI Service	Weeks 6-8	Flask app, liveness detection, pricing, matching	Completed
Phase 4: Mobile Apps	Weeks 9-12	React Native apps, authentication, booking UI	Pending
Phase 5: Dashboard	Weeks 16-17	React.js dashboard, monitoring, analytics	Pending
Phase 6: Safety	Weeks 13-15	GPS tracking, Safety Sentinel, alerts	Completed
Phase 7: AI Training	Weeks 18-19	Train models, K-Means clustering	Pending
Phase 8: Testing	Weeks 20-22	Unit, integration, UAT, deployment	Pending

5.3 Resource Allocation

Resource Type	Description	Quantity/Cost
Human Resources	Developers (Full-time)	2
Human Resources	Project Advisor (Part-time)	1
Technical Resources	Development laptops	2
Technical Resources	Cloud hosting (AWS/DigitalOcean)	PKR 10,000/month

Technical Resources	API costs (Google Maps, Twilio)	PKR 5,000/month
Technical Resources	Testing devices (Android/iOS)	PKR 50,000 (one-time)
Budget	Total Estimated Budget	PKR 140,000

6. APPENDICES

6.1 Glossary

Liveness Detection: Biometric technique to determine if a face is from a live person or a photograph/video.

Point-to-Line Distance: Geometric calculation measuring shortest distance between a point and a line segment.

Weighted Algorithm: Scoring method that assigns different importance levels to multiple factors.

Fairness Metric: Quantitative measure ensuring equitable distribution of resources.

6.2 AI Model Specifications

Model	Algorithm	Features/Architecture	Expected Performance
Pink Pass CNN	Convolutional Neural Network	3 conv layers + 2 dense layers	>98% accuracy, <5% FPR
Price Prediction	Linear Regression	7 features (distance, time, demand, etc.)	R2 >0.85
Driver Matching	Weighted Scoring	50% Distance + 30% Rating + 20% Fairness	N/A

6.3 Technology Stack

Component	Technology	Version
Backend	Node.js	v18+
Backend	Express.js	v4.18
Backend	MongoDB	v6.0
Backend	Socket.io	v4.7
AI Service	Python	v3.9+
AI Service	Flask	v3.1
AI Service	TensorFlow/Keras	v2.20
AI Service	OpenCV	v4.13
AI Service	scikit-learn	v1.8
Mobile Apps	React Native	v0.72+
Admin Dashboard	React.js	v18+

Document Approval

This Software Requirements Specification has been reviewed and approved by:

Project Advisor:
Tanveer Ahmed
Date: _____

Team Members:
Mirza Umer Ikram (S3F22UBSCS081)
Date: _____

Ruhma Bilal (S3F22UBSCS088)
Date: _____